



~~~ Sea otter foraging analysis (SOFA) V. 3.2 ~~~

Project: SOFA_input_patches_updated, Results file:
Rslt_All_2025_Oct_31_10hr.rdata

SOFA created for U.S. Geological Survey and Seattle Aquarium by M.T. Tinker

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Summary

The analytic approach is collectively referred to as the “Sea Otter Foraging Analysis”, or SOFA. Standard variables recorded in the field from foraging sea otters - duration of dive and surface intervals, prey captures, prey sizes, etc. - are first summarized for all the dives in each feeding bout, and then Bayesian methods are used to fit a process model to these observed data, in order to estimate key “latent” parameters. Latent parameters of interest include how sea otters allocate their effort to foraging for different prey types, how much each of these prey types contributes to the resulting diet, several prey-specific parameters (mean size, handling time, capture rates, and the functional relationship between the latter two parameters and prey size), and the overall net rate of biomass consumption and energy intake. The process model uses a probabilistic approach to account for incomplete data (including un-identified prey and missing data fields from some records), and the inherent biases associated with incomplete data (e.g. which types and sizes of prey are more likely to be recorded as un-identified). The resulting parameter estimates account for all sources of uncertainty, including sampling error, measurement error, uncertainty in the functional relationship between prey size and edible biomass, error in caloric density estimates, and various other sources of parameter uncertainty.

Methods

Observation model

SOFA is based around a simple conceptual model of sea otter foraging that corresponds to what an observer records in the field. Specifically, during a period of feeding activity (a “Bout”, consisting of a contiguous sequence of feeding dives), sea otters make decisions as to how to allocate their effort among multiple potential prey types. The term “prey type” is used here in a flexible way: a prey type may be a single species (*Tegula brunnea*), or it may be a group of related species (e.g. “marine snails”). Each prey type can be defined by

several observable metrics including its capture rate (the number of items encountered and captured per minute of time searching for that prey during dives), the time required to handle an item of that prey type once captured, the size of each item and the correlations between prey size and the capture rate and per-item handling time. The size of each captured prey item is recorded in terms of the maximum linear dimension relative to paw size, for later conversion to an absolute value (SZ_j , in mm). The total time in a bout allocated to each prey type j consists of the sum of the dive durations (DT_j) for dives allocated to acquiring that prey type, and the sum of time at the surface spent handling items of that prey type (HT_j), both of which are measured in seconds. For dives where multiple prey types are captured, we pro-rate the time among prey types: that is, the relevant DT and HT values are divided among prey types, proportional to their size and number. In addition to the confirmed time allocated to each prey type, there is also “unallocated time” (UT) during a bout, which consists of the total duration of unsuccessful dives and time at surface (ST) not handling prey. We can partition this unallocated time among prey types according to their proportional contributions to confirmed allocated time, PA_j . Thus, the total number of minutes (TM) allocated to prey type j in bout i is calculated as:

$$TM_{j,i} = \frac{1}{60} \left[\sum DT_{j,i} + \sum HT_{j,i} + \left(PA_{j,i} \sum UT_i \right) \right]$$

We note that one of the prey types for which we calculate total allocated minutes consists of un-identified prey items (UNID): we assume that these UNID prey items are a collection of all the other known prey types, but we do not know *a priori* the proportion of each known prey type comprising the UNID category (these values are to be estimated by the model, as explained below).

For each observed bout we calculate the total number of minutes allocated to each prey type ($TM_{j,i}$), and the mean value (averaged across dives) of four other statistics: the size of items of type j ($SZ_{obs,j,i}$), the handling time per item of prey type j ($H_{obs,j,i}$), the proportion of dives allocated to a prey type that are successful (PSD_j), and the per-item capture rate during dives allocated to prey type j ($cp_{obs,j,i}$). For the latter statistic we exclude handling time at the surface, thus ensuring that the capture rate statistic is equivalent to the *per-capita* attack rate parameter in type-II functional response models.

Process model

The observed activity of sea otter foraging can be approximated by a sequence of mathematical equations that together represent the process model, the expected dynamics of which are determined by the values of the parameters in the equations (Table 1). We let η_j represent the mean proportional allocation of foraging effort to prey type j , excluding the UNID class (i.e. TRUE effort allocation if all prey were positively identified), such that:

$$\sum_{j=1}^J \eta_j = 1$$

For each prey type j we also specify parameter ω_j as the probability that an item of that prey type may not be positively identified. We calculate values of ω_j based on the empirical distributions of the log of handling time and the log of mean prey size of prey type j , and the degree to which these distributions overlap with the same distributions for the UNID prey class. We measure joint proportional overlap of multiple distributions using the Bhattacharyya distance metric (BC_j), calculating ω_j as $\exp(-BC_j)$. This approach reflects the assumption that the more similar the joint density distributions of size and handling time between UNID and prey type j , the more likely it is that j contributes to the UNID prey class. To account for unidentified prey in our observed data set, we define the parameter α as the relative allocation of effort to each prey type INCLUDING the UNID prey class. For positively identified prey types:

$$\alpha_j = \eta_j \cdot (1 - \omega_j) \cdot \tau_B$$

while for the unidentified prey class (UNID):

$$\alpha_u = \sum_j \eta_j \cdot \omega_j \cdot \tau_B$$

In the above equations, parameter τ_B represents a fitted precision parameter, allowing us to use α_j as the base parameters for a Dirichlet distribution that defines the relative probabilities of a prey type being observed in a given bout:

$$[\theta_{j,i}] \sim \text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_J, \alpha_u)$$

where $\theta_{j,i}$ is the expected proportional allocation of effort to each prey type for bout i .

We define parameter $\mu_{s,j}$ as the mean log size (mm) for each prey type. For handling time and capture rate, we note that both of these parameters are correlated strongly with prey size: specifically, there is an approximately linear relationship between the log of each variable and the log of prey size. We therefore calculate expected log handling time ($\mu_{h,j}$) and expected log capture rate ($\mu_{c,j}$) as derived parameters given the size of prey type j observed on a given bout:

$$\mu_{h,j} = \psi_{1,j} + \psi_{2,j} \cdot \log(SZ.obs_j)$$

$$\mu_{c,j} = \phi_{1,j} - \phi_{2,j} \cdot \log(SZ.obs_j)$$

where the fitted parameters $\phi_{1,j}$, $\phi_{2,j}$, $\psi_{1,j}$, and $\psi_{2,j}$, together describe the functional relationships between handling time, capture rate, and prey size for each prey type. We allow for variation in log size, log handling time and log capture rate across bouts by defining variance parameters $\sigma_{s,j}$, $\sigma_{h,j}$, and $\sigma_{c,j}$. We can also calculate mean parameter values averaged over all bouts: specifically, if we define $\bar{\mu}_{s,j}$ as the mean log size of prey type j over the entire data set, then we can calculate mean size, handling time and capture rate for prey type j as:

$$\bar{S}_j = \exp\left(\bar{\mu}_{s,j} + \sigma_{s,j}^2/2\right)$$

$$\bar{H}_j = \exp\left((\mu_{h,j}|\bar{\mu}_{s,j}, \psi_j) + \sigma_{h,j}^2/2\right)$$

$$\bar{cp}_j = \exp\left((\mu_{c,j}|\bar{\mu}_{s,j}, \phi_j) + \sigma_{c,j}^2/2\right)$$

We define parameter λ_j the expected proportion of successful dives associated with each prey type, which we estimate as a logit parameter with Cauchy prior:

$$\text{logit}(\lambda_j) \sim \text{Cauchy}(0, 2.5)$$

And the overall mean dive success rate is calculated as:

$$\bar{\lambda} = \sum_{j=1}^J \eta_j \cdot \bar{\lambda}_j$$

We next define several “derived” parameters that help simplify or expand our interpretation of model results. The biomass consumption rate for prey type j (CR_j) during foraging time allocated to that prey type can be calculated using Hollings disc equation (i.e. the type-II function response equation):

$$CR_j = \frac{cp_j \cdot m_j}{1 + cp_j \cdot H_j}$$

where the per-item biomass of prey type j (m_j) is calculated from taxa-specific log-log relationships between maximum linear dimension (mm) and wet edible biomass (g), based on published data. We note that the handling time, capture rate and biomass values are all assumed to vary across bouts, corresponding to variation in prey size: we account for this source of uncertainty, as well as the uncertainty in the fitted mass-length relationships, by using a re-sampling approach in the calculation of CR_j . We calculate the prey-specific energy intake rates in a similar way:

$$ER_j = \frac{cp_j \cdot m_j \cdot Cdens_j}{1 + cp_j \cdot H_j}$$

where $Cdens_j$ is the caloric density (kcal/g) of items of prey type j , based on published data.

In addition to prey-specific parameters, we also integrate consumption rate and energy intake rates across all prey types, accounting for proportional allocation of effort among prey types, to obtain the overall consumption rate (CR) and energy intake rate (ER):

$$\overline{CR} = \sum_j \eta_j \cdot CR_j$$

$$\overline{ER} = \sum_j \eta_j \cdot ER_j$$

where η_j represents the mean proportional allocation of foraging effort to prey type j , as defined above.

Diet composition, defined as the proportional contribution (in terms of consumed biomass) of each prey type to the overall diet ($/pi_j$), is calculated as:

$$\pi_j = (\eta_j \cdot \overline{CR}_j) / \sum_{j=1}^J \eta_j \cdot \overline{CR}_j$$

The proportional contribution of each prey type to the UNID prey class is represented by parameter v_j , calculated as:

$$v_j = \omega_j \cdot \pi_j \cdot 1/EB_j$$

where EB_j is the average biomass per prey item of prey type j .

Finally, the process model can be modified to account for random effects of categorical group variables (age, sex, area, time period) by utilizing a hierarchical approach for certain key parameters. We allow foraging effort to vary across groups using a Dirichlet-Multinomial approach:

$$\eta_{g,j} \sim \text{Dirichlet}(\eta_j \cdot \tau_G)$$

where $\eta_{g,j}$ is the mean proportional allocation of foraging effort to prey type j in bouts belonging to group level g , and parameter τ_G is a fitted precision parameter that determines the degree of consistency in diet across groups. We assume that log prey size for each prey type is normally distributed across groups with mean equal to $\bar{\mu}_{s,j}$ and standard error as a fitted parameter. We make the same assumption for $\bar{\phi}_{1,j}$, $\bar{\psi}_{1,j}$ and $\bar{\lambda}_j$, thereby allowing prey specific handling times, capture rates and dive success rates to vary across groups. By treating these base parameters hierarchically, we also allow for variation in the derived parameters of diet composition, mean consumption rates and mean energy intake rates across groups. Table 1 provides a summary of all parameters estimated by the model.

Table 1. Summary of estimated parameters

Parameter	Description
$\overline{CR}$	Mean overall net biomass consumption rate (CR, g/min) while foraging
$\overline{ER}$	Mean overall net energy intake rate (ER, kcal/min) while foraging
$\bar{\lambda}$	Mean overall dive success rate (proportion successful dives)
$\bar{S}_j$	Mean size, prey type j
$\bar{H}_j$	Mean handling time, prey type j
$\bar{c}p_j$	Mean capture rate, prey type j
$\overline{CR}_j$	Mean biomass consumption rate, prey type j
$\overline{ER}_j$	Mean energy intake rate, prey type j
$\bar{\lambda}_j$	Mean dive success rate, prey type j

Parameter	Description
$\bar{\phi}_{1,j}$	cp vs log(Size) function, intercept parameter, prey type j
$\phi_{2,j}$	cp vs log(Size) function, slope parameter, prey type j
$\bar{\psi}_{1,j}$	H vs log(Size) function, intercept parameter, prey type j
$\psi_{2,j}$	H vs log(Size) function, slope parameter, prey type j
$\bar{\eta}_j$	Proportion of foraging effort allocated to prey type j
$\bar{\pi}_j$	Proportion of diet (biomass consumed) made up of prey type j
$\bar{\omega}_j$	Proportion of prey type j identified (not recorded as “un-identified” prey)
$\bar{v}_j$	Proportional contribution of prey type j to un-identified prey
$\sigma_{c,j}$	Std error in log(CR) across bouts for a given prey type
$\sigma_{h,j}$	Std error in log(H) across bouts for a given prey type
$\sigma_{s,j}$	Std error in log(S) across bouts for a given prey type
$\sigma_{l,j}$	Std error in logit(lambda) across bouts for a given prey type
τ_B	Precision (consistency) in diet composition across bouts (within group)
τ_G	Precision (consistency) in diet composition across groups (if defined)
CR_g	Mean net consumption rate (CR, g/min) while foraging, group g
ER_g	Mean net energy intake rate (ER, kcal/min) while foraging, group g
$\bar{\lambda}_g$	Mean overall dive success rate, group g
$S_{g,j}$	Mean size, prey type j, group g
$H_{g,j}$	Mean handling time, prey type j, group g
$cp_{g,j}$	Mean capture rate, prey type j, group g
$CR_{g,j}$	Mean consumption rate, prey type j, group g
$ER_{g,j}$	Mean energy intake rate, prey type j, group g
$\lambda_{g,j}$	Mean dive success rate, prey type j, group g
$\phi_{1,g,j}$	cp vs log(Size) function, intercept parameter, prey type j, group g
$\psi_{1,g,j}$	H vs log(Size) function, intercept parameter, prey type j, group g
$\eta_{g,j}$	Proportion of foraging effort allocated to prey type j, group g
$\pi_{g,j}$	Proportion of diet (biomass consumed) made up of prey type j, group g
$\omega_{g,j}$	Proportion of prey type j un-identified, group g
$v_{g,j}$	Contribution of prey type j to un-identified prey, group g

Note: parameters with ‘g’ subscripts estimated if by-groups were incorporated in analysis

Relating observation model and process model

By comparing expected distributions from the process model with observed data, the statistics recorded from foraging bouts constrain the possible values of the parameters of the process model. Specifically, we assume that the observed distribution of minutes allocated to each prey type on a given bout can be described by a multinomial distribution:

$$[TM_{j,i}] \sim \text{Multinomial}([\theta_{j,i}])$$

We assume that observed mean prey size for prey type j on bout i is described by a log-normal distribution:

$$SZ.obs_{j,i} \sim \text{lognormal}(\mu_{s,j}, \sigma_{s,j})$$

where $\sigma_{s,j}$ is a parameter describing the variance in the mean size of prey j across bouts.

We assume that observed mean handling time and mean capture rate for prey type j on bout i are also described by log-normal distributions:

$$H.obs_{j,i} \sim \text{lognormal}(\psi_{1,j} + \psi_{2,j} \cdot \log(SZ.obs_{j,i}), \sigma_{h,j})$$

$$cp.obs_{j,i} \sim \text{lognormal}(\phi_{1,j} - \phi_{2,j} \cdot \log(SZ.obs_{j,i}), \sigma_{c,j})$$

where $\sigma_{h,j}$ and $\sigma_{c,j}$ are fitted parameters describing variance in these statistics across bouts.

We assume that the observed dive success rates specific to each prey type (PSD_j), logit-transformed, are described by a normal distribution:

$$\text{logit}(PSD_j) \sim \text{normal}(\text{logit}(\lambda_j), \sigma_{l,j})$$

where $\sigma_{l,j}$ is a fitted parameter describing variance in logit dive success rate across bouts.

We use standard Markov-Chain Monte Carlo methods to fit the model to the foraging data, with uninformative priors for all model parameters (Cauchy priors for unconstrained parameters and half-Cauchy priors for parameters constrained to be positive). We evaluate model convergence by graphically examining chain mixing and ensuring that the Rhat statistic is close to 1 for all estimated parameters. We evaluate model fit using graphical posterior predictive checks, ensuring that the distributions of out-of-sample predictions are consistent with observed data. We present summaries of posterior distributions for both base parameters and derived parameters such as biomass consumption and energy intake rate.

Results

Both graphical and tabular results are presented below. In some cases prey types are referred to numerically (e.g. as subscripts for prey-specific parameters), in which case the numbers correspond to prey types as summarized in Table 2.

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

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```

Table 2: Prey types included in the analysis

TypeN	PreyType	Description	Class	Min_size_mm
1	urchin_b	urchins, various sp	urchin	NA
2	urchin_f	urchins, various sp	urchin	NA
3	urchin_i	urchins, various sp	urchin	NA
4	red_urchin	red urchins	urchin	NA
5	mussel	mussels	mussel	NA
6	clam	clams, various species	clam	NA
7	cancrid_crab	Cancr crabs	cancrid_crab	NA
8	kelp_crab	kelp crabs	kelp_crab	NA
9	crab_other	Other crabs	other_crab	NA
10	snail	snails, various sp	snail	NA
11	other	chitons, limpets, barnacles,etc.	other_hardsub	NA
12	worm	fat innkeepers, nereids, etc.	worm	NA
13	UNID	UN-IDENTIFIED	NA	NA

The posterior estimates for net consumption rate (CR) and energy intake rate (ER), for the data set as a whole, are shown in Figure 1, and summarized in Table 3. Posterior density plots are also shown for estimates of foraging effort allocation among prey types, proportional contribution to diet (in terms of consumed biomass) by prey type, and estimates mean handling time, size, consumption rate and energy intake rate for each prey type.

If the SOFA analysis being summarized incorporated group-level differences in foraging behavior (e.g. area-based differences, time-based differences or differences among individual animals), a second series of plots are presented showing the same statistics described above but for each level of the grouping variable(s).

Tabular summaries of all statistics (both for all data combined and by group levels, if appropriate) are provided at the end of the report.

Figures, statistics for ALL data

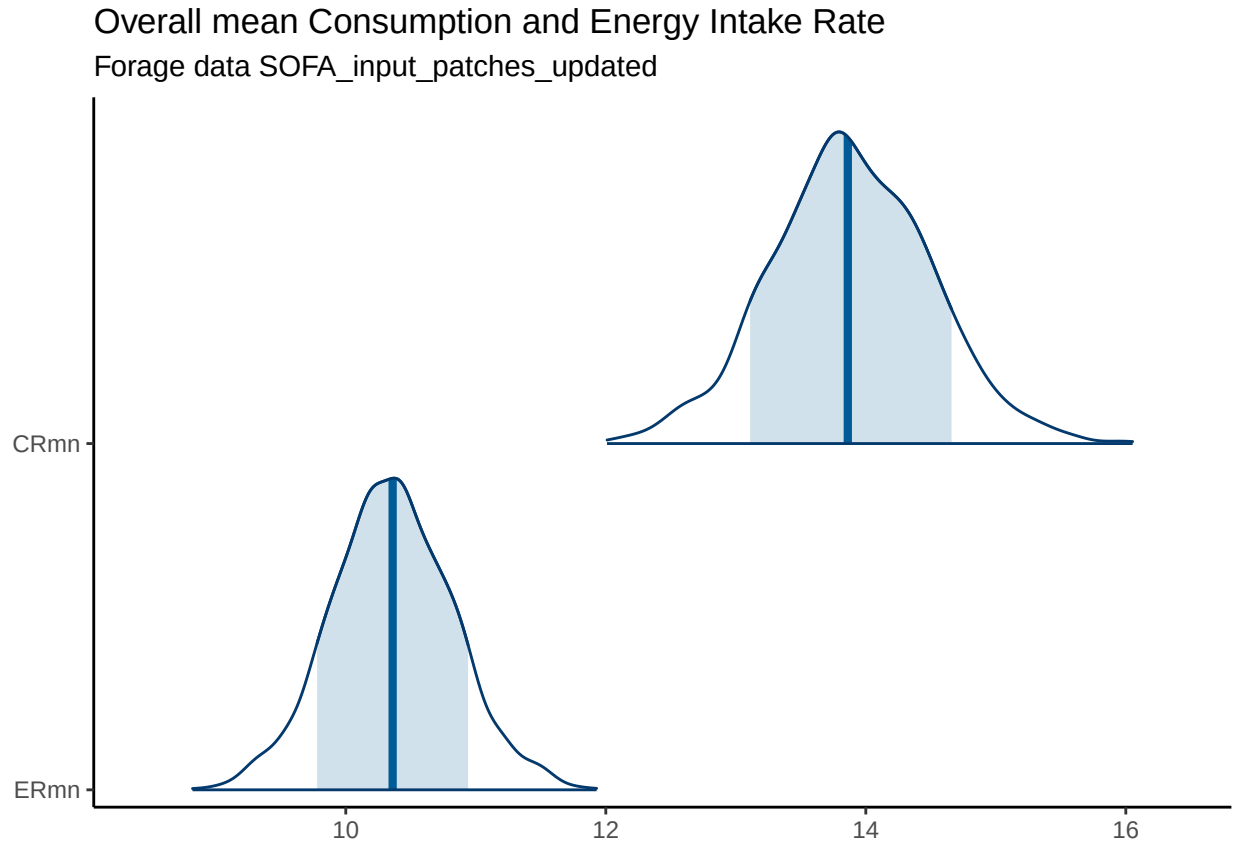


Figure 1: Density plot showing posterior distributions for consumption rate ($\bar{C}R$, g/min) and rate of energy intake ($\bar{E}R$, kcal/min) for the overall data set

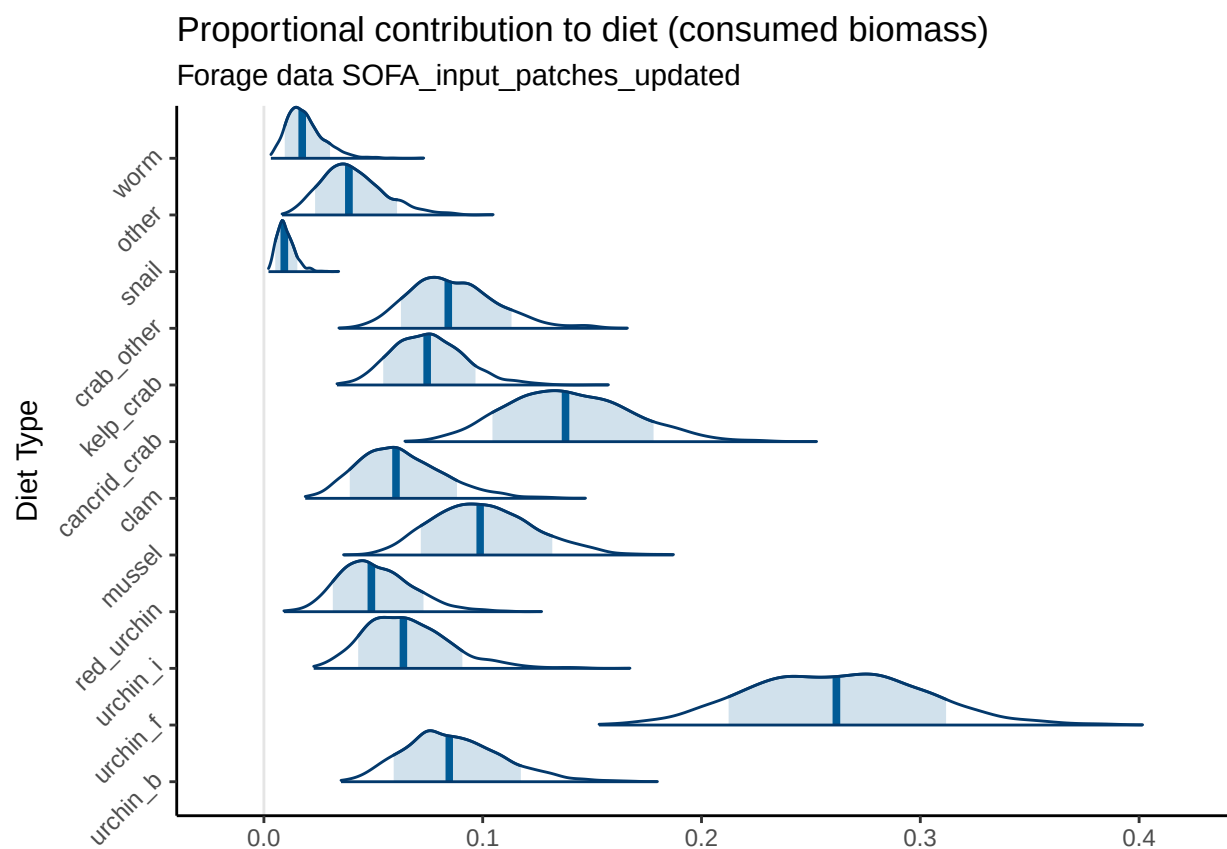


Figure 2: Density plot showing posterior distributions for π_j , the proportion of diet (biomass consumed) made up of prey type j

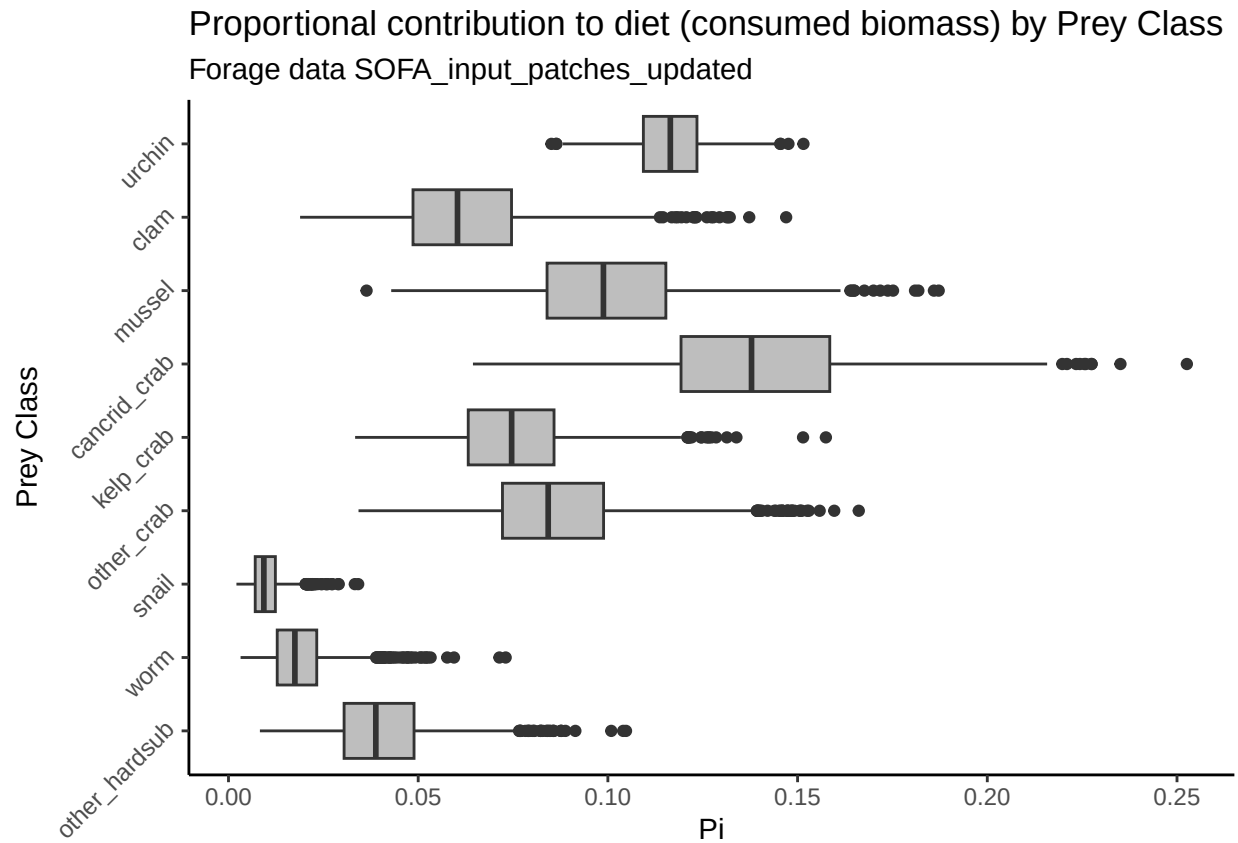


Figure 3: Boxplot showing posterior distributions for diet composition by Prey Class

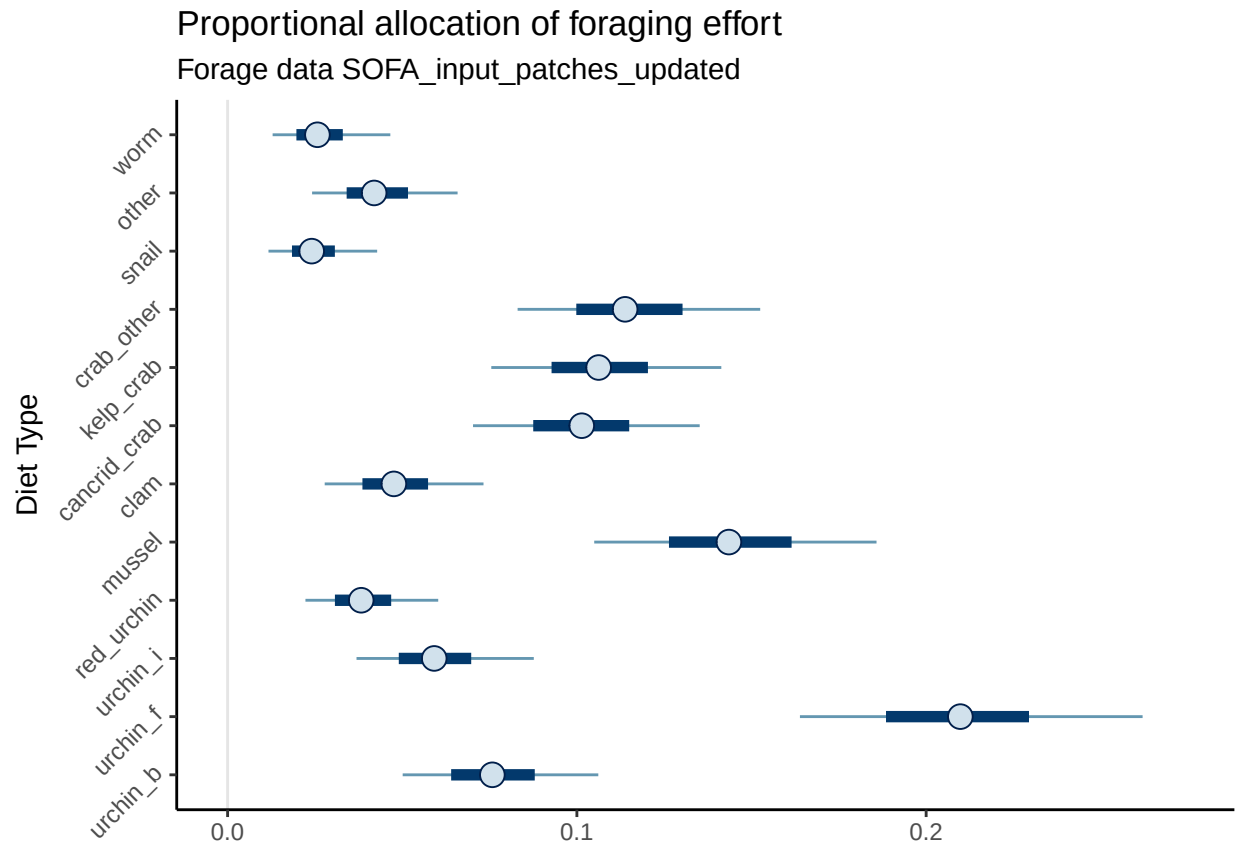


Figure 4: Caterpillar plot showing posterior distributions for η_j , the relative allocation of effort to each prey type j

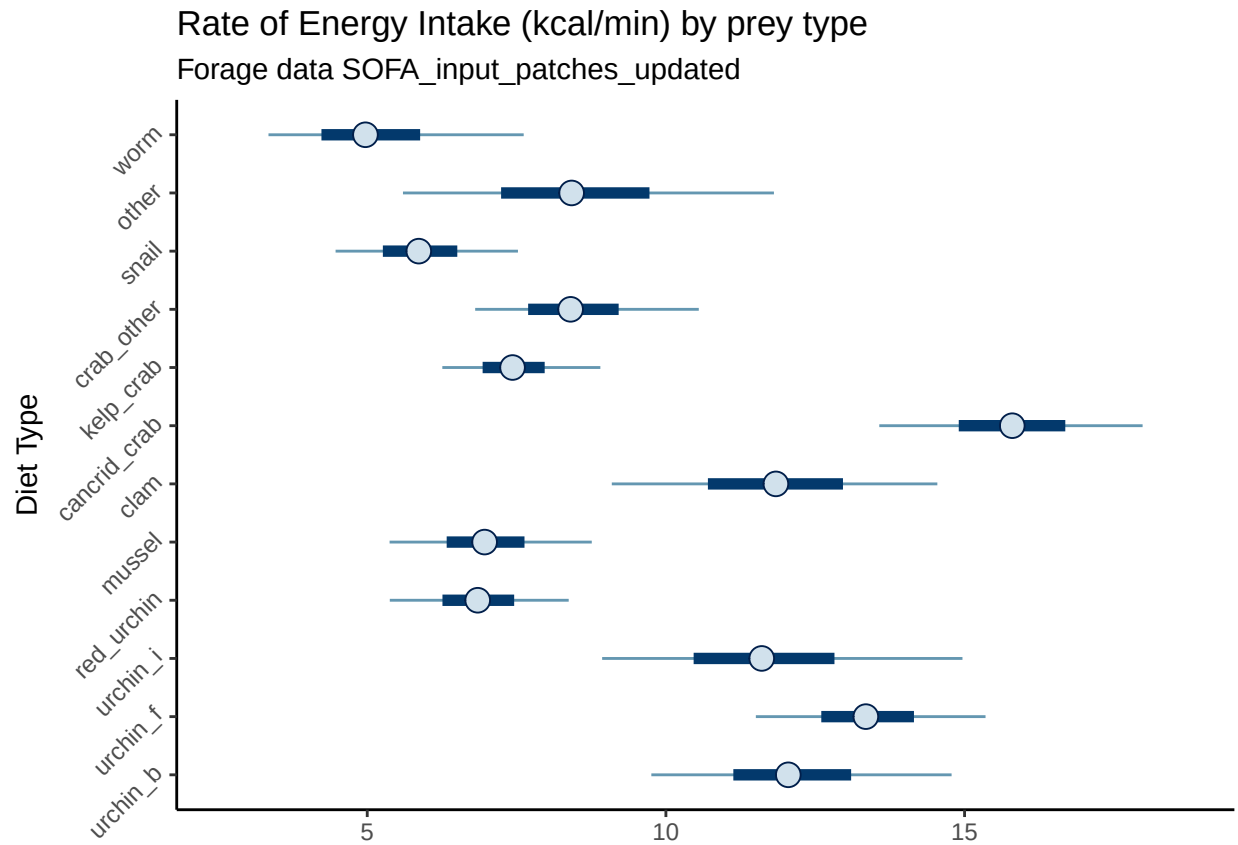


Figure 5: Caterpillar plot showing posterior distributions for the rate of energy intake while feeding on each prey type j

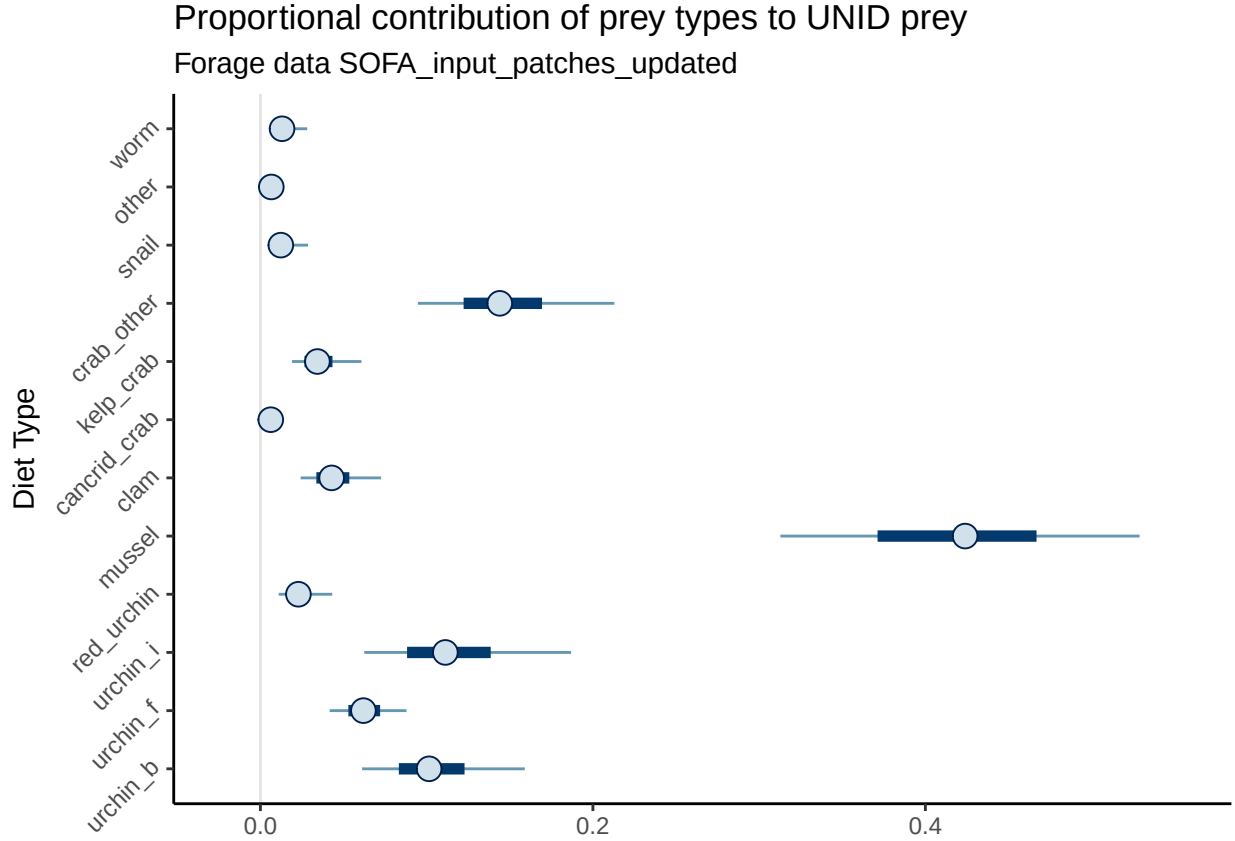


Figure 6: Caterpillar plot showing posterior distributions for v_j , the relative contribution to un-identified of each prey type j

Tables, statistics for ALL data

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## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
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Table 3: Parameter estimates for consumption rate ($\bar{C}R$, g/min), rate of energy intake ($\bar{E}R$, kcal/min), and dive success rate ($\bar{\lambda}$) for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
CR_bar	13.8800	0.6133	12.6200	15.1100	1767	1.001
ER_bar	10.3600	0.4660	9.4170	11.3000	1762	1.001
Lambda_bar	0.8588	0.0279	0.7974	0.9059	1649	1.002

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```

Table 4: Parameter estimates for mean size, $\bar{S}_j$ (mm), by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
S_bar_1	47.71063	1.7693181	44.31280	51.41839	1822.745	0.9998590
S_bar_2	47.81796	1.1739761	45.51570	50.23781	1662.920	1.0055286
S_bar_3	46.73529	2.5073874	42.12324	52.11586	1503.156	1.0015064
S_bar_4	72.84666	5.6682765	62.63774	84.94629	1639.094	0.9995456
S_bar_5	48.17103	1.5060246	45.24587	51.30234	1590.691	1.0007384
S_bar_6	64.29661	5.8187390	54.48033	77.24614	1622.602	1.0008610
S_bar_7	105.29888	7.1050904	92.42786	120.65940	1718.500	0.9985272
S_bar_8	44.60317	2.4574828	40.31569	49.99497	1625.603	1.0008160
S_bar_9	54.20709	3.3910959	48.22460	61.62439	1645.016	1.0038499
S_bar_10	23.34098	0.6342194	22.12426	24.77253	1695.026	1.0012945
S_bar_11	97.76782	22.9399759	66.50924	155.14517	1650.603	1.0017538
S_bar_12	154.94512	42.1911086	100.98730	252.31985	1601.034	1.0006838

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Table 5: Parameter estimates for mean handling time, $\bar{H}_j$ (sec), by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
H_bar_1	29.17247	3.386399	23.811128	36.91376	1546.220	1.001216
H_bar_2	34.02815	3.236048	28.748520	41.56951	1668.882	1.003687
H_bar_3	28.54236	3.067925	23.674788	35.44775	1607.770	1.000519
H_bar_4	60.43235	10.691025	43.778395	86.61800	1805.379	1.001304
H_bar_5	20.21454	2.079539	16.781543	24.71545	1769.340	1.001256
H_bar_6	42.66524	7.475000	31.272227	61.24210	1653.704	1.003699
H_bar_7	143.19644	17.099040	113.402275	180.09033	1630.006	1.002496
H_bar_8	40.14642	6.387032	30.774175	55.86290	1569.918	1.001942
H_bar_9	41.07820	3.505618	34.827678	48.72219	1540.810	1.000312
H_bar_10	14.49880	3.847769	9.496593	24.38255	1503.236	1.002393
H_bar_11	90.16898	26.326871	51.689805	154.79240	1631.071	1.000708
H_bar_12	44.73044	7.646639	33.071750	61.70415	1440.701	1.003197

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Table 6: Parameter estimates for mean consumption rate by prey type $\bar{c}r_j$ (g/min), for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
cr_1	15.687252	1.923269	12.042388	19.740430	1682.074	0.9988666
cr_2	17.314038	1.495928	14.447677	20.405963	1693.410	1.0006212
cr_3	15.142686	2.401286	10.926208	20.381248	1594.309	1.0025105
cr_4	18.063132	2.393152	13.349282	22.841310	1739.630	0.9989814
cr_5	9.662965	1.411983	6.989954	12.608028	1653.480	1.0027698
cr_6	17.856849	2.473799	12.925648	22.750132	1630.815	1.0016104
cr_7	19.057405	1.647330	15.859908	22.268287	1668.645	1.0002568
cr_8	9.762388	1.075643	7.850518	12.215523	1496.956	1.0018390
cr_9	10.398888	1.371436	8.050626	13.356825	1662.854	1.0042547
cr_10	5.514831	0.878460	3.933676	7.286289	1723.152	1.0023858
cr_11	13.095896	2.855132	7.908065	19.414995	1693.258	1.0017439
cr_12	9.619477	2.462740	5.700871	15.328660	1634.500	1.0013214

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```

Table 7: Parameter estimates for mean energy intake rate by prey type $\bar{e}r_j$ (kcal/min), for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
er_1	12.138113	1.4946238	9.329149	15.186497	1659.939	0.9986156
er_2	13.394461	1.1606504	11.158225	15.754145	1666.828	1.0009552
er_3	11.718782	1.8639296	8.429861	15.826525	1589.031	1.0025216
er_4	6.864840	0.9116015	5.049670	8.696312	1735.241	0.9993094
er_5	6.991858	1.0267718	5.060402	9.107160	1639.294	1.0033048
er_6	11.840441	1.6471556	8.580470	15.069392	1637.246	1.0017203
er_7	15.807496	1.3698256	13.143982	18.508025	1676.227	1.0002349
er_8	7.488156	0.8287634	6.027312	9.318264	1548.913	1.0017658
er_9	8.490797	1.1275307	6.570061	10.943250	1668.587	1.0025769
er_10	5.912579	0.9450741	4.219292	7.845776	1712.335	1.0032692
er_11	8.517116	1.8587623	5.148778	12.526880	1686.680	1.0019842
er_12	5.142647	1.3166612	3.043491	8.197330	1634.626	1.0013185

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 8: Parameter estimates for λ_j , mean dive success rate by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
lambda_1	0.9467992	0.0168092	0.9054729	0.9707797	1959.850	0.9981449
lambda_2	0.9352719	0.0149430	0.9006427	0.9600917	1809.916	1.0086001
lambda_3	0.9440152	0.0333728	0.8572829	0.9820810	1616.002	1.0022739
lambda_4	0.8102155	0.1938039	0.2656732	0.9891471	1715.815	0.9996152
lambda_5	0.9604832	0.0099054	0.9375700	0.9761463	1750.736	1.0016627
lambda_6	0.7876061	0.2006816	0.2309059	0.9879630	1735.832	1.0061094
lambda_7	0.4902501	0.1356033	0.2313227	0.7838434	1499.011	1.0012658
lambda_8	0.8893911	0.0608935	0.7267669	0.9674757	1599.661	0.9992536
lambda_9	0.8548939	0.1123126	0.5350365	0.9750892	1714.550	1.0009517
lambda_10	0.9742118	0.0159248	0.9357682	0.9902930	1411.247	1.0061895
lambda_11	0.8912799	0.1798998	0.2764463	0.9954443	1547.210	1.0015148
lambda_12	0.6156858	0.1389724	0.2996227	0.8749947	1612.486	1.0010903

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 9: Estimates for $\phi_{1,j}$, the intercept parameter for the function relating log consumption rate to log size, by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
phi_1_1	0.7316659	0.3721044	0.1928483	1.6353213	1714.647	1.0035677
phi_1_2	0.9472516	0.5064930	0.2988333	2.2215117	1454.423	1.0027305
phi_1_3	0.5802488	0.2619104	0.1910058	1.2306895	1291.869	0.9992426
phi_1_4	0.0913333	0.4469338	-0.5463502	1.2704807	1622.770	1.0004293
phi_1_5	1.3016739	0.4370671	0.5999867	2.3964502	1394.606	1.0016270
phi_1_6	0.8610716	0.5915249	-0.1615909	2.0692708	1395.313	1.0032343
phi_1_7	-0.4485603	0.1643485	-0.7165807	-0.0527529	1647.806	1.0000255
phi_1_8	0.5590944	0.3323165	-0.0159267	1.2301375	1411.742	1.0035784
phi_1_9	0.1456047	0.2290171	-0.2400716	0.6447913	1480.946	0.9990704
phi_1_10	1.6922491	0.3481522	1.0316050	2.4173072	1557.002	1.0047269
phi_1_11	-0.4378142	0.2759861	-0.8900276	0.2030198	1443.178	0.9992640
phi_1_12	-0.5276262	0.3310318	-1.0326110	0.3186229	1392.401	1.0056081

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```



```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 10: Estimates for $\phi_{2,j}$, the slope parameter for the function relating log consumption rate to log size, by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
phi_2_1	0.1420132	0.1326399	0.0044856	0.4913645	1526.341	1.0054079
phi_2_2	0.2219120	0.1899872	0.0085930	0.6873408	1492.406	1.0030084
phi_2_3	0.0946711	0.0938010	0.0026734	0.3477689	1275.272	0.9982273
phi_2_4	0.1186799	0.1077295	0.0049678	0.4103549	1536.444	1.0011360
phi_2_5	0.1169245	0.1354118	0.0029734	0.5080111	1431.047	1.0039885
phi_2_6	0.3037728	0.1679646	0.0238085	0.6526834	1379.604	1.0044352
phi_2_7	0.0366379	0.0310672	0.0014313	0.1182545	1332.950	1.0066064
phi_2_8	0.2214590	0.1183438	0.0218770	0.4668609	1436.365	1.0025218
phi_2_9	0.1200941	0.0762364	0.0103116	0.2977377	1411.557	1.0013016
phi_2_10	0.2935416	0.1854586	0.0213743	0.6986700	1505.670	1.0026400
phi_2_11	0.0815872	0.0689813	0.0038460	0.2579213	1471.274	0.9981304
phi_2_12	0.0589855	0.0510025	0.0019985	0.1950825	1476.248	1.0052807

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 11: Estimates for $\psi_{1,j}$, the intercept parameter for the function relating log handling time to log size, by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
psi_1_1	2.628020	0.5087644	1.5181842	3.376076	1613.959	1.0006238
psi_1_2	2.149610	0.5794875	1.0770642	3.284780	1678.968	0.9990456
psi_1_3	2.663682	0.4056195	1.7914525	3.257377	1492.797	1.0016500
psi_1_4	2.189507	0.8477478	0.8218658	3.894034	1390.192	1.0025738
psi_1_5	2.296741	0.4251369	1.3965995	2.957824	1611.915	0.9998790
psi_1_6	2.898158	0.4829950	1.7948002	3.624736	1630.764	1.0023479
psi_1_7	2.617667	0.3678658	1.9010400	3.346859	1578.699	1.0016397
psi_1_8	1.119317	0.2345549	0.7149761	1.635174	1241.639	1.0073300
psi_1_9	3.295330	0.1945620	2.8589110	3.614755	1290.452	1.0027328
psi_1_10	1.933003	0.3070380	1.3654743	2.579155	1471.503	1.0010280
psi_1_11	2.329572	0.3883447	1.5772920	3.087934	1515.131	1.0012089
psi_1_12	2.883479	0.4390038	1.9152113	3.682812	1386.574	0.9990225

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 12: Estimates for $\psi_{2,j}$, the slope parameter for the function relating log handling time to log size, by prey type, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
psi_2_1	0.2616391	0.1827278	0.0099060	0.6612842	1601.116	1.0011180
psi_2_2	0.4832732	0.2201208	0.0479295	0.8939022	1706.447	0.9993294
psi_2_3	0.2388295	0.1625192	0.0090854	0.5967996	1398.979	1.0023059
psi_2_4	0.4809135	0.2163217	0.0445491	0.8358176	1371.636	1.0012749
psi_2_5	0.2392077	0.1522747	0.0170226	0.5729597	1620.483	0.9999543
psi_2_6	0.2053248	0.1393421	0.0124594	0.5272689	1593.329	1.0024145
psi_2_7	0.4657876	0.0813488	0.3053923	0.6265992	1586.382	1.0008385
psi_2_8	1.0078447	0.0945004	0.8036957	1.1651255	1237.830	1.0065162
psi_2_9	0.1051116	0.0658303	0.0057337	0.2584635	1153.985	1.0046962
psi_2_10	0.5673020	0.1794880	0.1456259	0.8894697	1320.796	1.0007326
psi_2_11	0.4528496	0.0935225	0.2693323	0.6291796	1528.613	1.0011726
psi_2_12	0.1615297	0.0785475	0.0226877	0.3337538	1385.246	0.9999897

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 13: Estimates for η_j , proportional allocation of effort to prey type j , for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
eta_1	0.0768201	0.0175475	0.0459639	0.1139041	1720.305	0.9996257
eta_2	0.2104188	0.0301192	0.1542618	0.2715169	1786.928	0.9992536
eta_3	0.0602498	0.0155695	0.0328203	0.0949044	1755.216	1.0022295
eta_4	0.0393481	0.0120137	0.0184020	0.0653675	1796.475	1.0015126
eta_5	0.1443783	0.0255247	0.0986747	0.1982557	1792.512	1.0009812
eta_6	0.0488017	0.0141302	0.0245807	0.0809181	1776.830	1.0019520
eta_7	0.1018978	0.0198881	0.0668711	0.1426916	1853.857	0.9999753
eta_8	0.1069688	0.0201861	0.0705505	0.1482353	1628.277	1.0013666
eta_9	0.1154640	0.0215308	0.0775408	0.1611200	1826.173	1.0005932
eta_10	0.0250762	0.0093579	0.0101859	0.0463791	1790.263	1.0013763
eta_11	0.0433065	0.0130489	0.0208302	0.0712942	1794.261	0.9991291
eta_12	0.0272698	0.0104525	0.0112842	0.0525022	1765.887	1.0032934

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 14: Estimates for π_j , proportion of diet (consumed biomass) consisting of prey type j, for the overall data set

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
pi_1	0.0868352	0.0224100	0.0490164	0.1350159	1714.584	1.0001363
pi_2	0.2620036	0.0393866	0.1897783	0.3418947	1840.169	0.9998943
pi_3	0.0656678	0.0197794	0.0337807	0.1106799	1619.130	1.0037455
pi_4	0.0511333	0.0166877	0.0234477	0.0882968	1842.043	1.0015043
pi_5	0.1005564	0.0229542	0.0616381	0.1491411	1795.193	1.0008700
pi_6	0.0626444	0.0196383	0.0310116	0.1073619	1733.430	1.0022616
pi_7	0.1396674	0.0282001	0.0898494	0.1972476	1680.493	1.0024298
pi_8	0.0753354	0.0166464	0.0463080	0.1123017	1700.540	1.0033160
pi_9	0.0866361	0.0201855	0.0530300	0.1320996	1670.827	0.9994177
pi_10	0.0099986	0.0041864	0.0037468	0.0203851	1730.814	1.0033884
pi_11	0.0406365	0.0145123	0.0172380	0.0739618	1752.411	1.0024282
pi_12	0.0188853	0.0086673	0.0059652	0.0389636	1840.713	1.0002192

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 15: Estimates for ω_j , probability that prey type j is positively identified (and thus not recorded as 'Un-ID' prey), and v_j , relative contribution of each prey type to the UNID prey category.

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
omega_1	0.2052473	0.0344174	0.1451538	0.2773235	1682.338	1.0012038
omega_2	0.2204333	0.0305453	0.1654747	0.2854033	1728.352	1.0026299
omega_3	0.2035237	0.0368066	0.1396519	0.2834081	1610.065	1.0021360
omega_4	0.1465541	0.0365266	0.0814472	0.2241954	1737.061	1.0014385
omega_5	0.2336840	0.0338475	0.1708501	0.3008847	1597.499	1.0014155
omega_6	0.2450200	0.0381941	0.1771975	0.3230127	1662.422	1.0013134
omega_7	0.0973049	0.0238160	0.0566728	0.1475232	1668.424	1.0009004
omega_8	0.0956675	0.0282286	0.0533945	0.1626504	1828.567	1.0006776
omega_9	0.2709203	0.0372444	0.2030851	0.3460099	1627.619	1.0011733
omega_10	0.0798090	0.0240053	0.0447062	0.1388722	1547.815	1.0000014
omega_11	0.1658019	0.0447478	0.0792759	0.2530611	1778.964	1.0013118
omega_12	0.1432876	0.0401371	0.0733985	0.2258194	1716.134	0.9981407
upsilon_1	0.1049063	0.0306358	0.0529631	0.1748110	1810.745	1.0007927
upsilon_2	0.0630943	0.0141678	0.0387732	0.0934990	1762.399	0.9983980
upsilon_3	0.1166430	0.0384119	0.0563450	0.2040725	1700.238	1.0020332

Table 15: (continued)

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
upsilon_4	0.0245474	0.0101146	0.0092529	0.0489804	1832.331	1.0012255
upsilon_5	0.4201977	0.0676010	0.2869464	0.5492908	1735.114	1.0020496
upsilon_6	0.0448436	0.0150641	0.0210207	0.0804455	1756.554	1.0010799
upsilon_7	0.0065041	0.0021789	0.0031451	0.0115471	1737.995	0.9996149
upsilon_8	0.0362112	0.0131130	0.0170436	0.0680718	1536.716	1.0036481
upsilon_9	0.1475457	0.0358366	0.0887288	0.2254538	1643.344	0.9993347
upsilon_10	0.0140762	0.0080590	0.0046726	0.0337493	1616.186	1.0005993
upsilon_11	0.0070411	0.0030066	0.0025378	0.0144377	1784.055	0.9995661
upsilon_12	0.0143894	0.0074271	0.0044310	0.0329862	1795.122	1.0015857

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 16: Estimates for model variance and precision parameters. Prey-specific standard error values are shown for log-normally distributed observed variables of prey size ($\sigma_{s,j}$), handling time ($\sigma_{h,j}$), consumption rate ($\sigma_{c,j}$) and dive success rate ($\sigma_{l,j}$). Also shown are precision parameters for Dirichlet distributions that describe the relative frequencies of different prey types. Precision parameters determine the consistency in diet composition across bouts (τ_b) and, if applicable, across different groups (τ_g)

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
sigma_s_1	0.15090	0.02780	0.10880	0.2135	1883	1.0020
sigma_s_2	0.16090	0.01734	0.13150	0.1989	1761	1.0030
sigma_s_3	0.13030	0.04201	0.07481	0.2371	1535	1.0020
sigma_s_4	0.27730	0.05360	0.19470	0.4035	1714	1.0000
sigma_s_5	0.17900	0.02214	0.14370	0.2305	1737	0.9995
sigma_s_6	0.36000	0.06151	0.26430	0.5062	1747	1.0040
sigma_s_7	0.44920	0.04618	0.37270	0.5498	1604	1.0010
sigma_s_8	0.15210	0.04067	0.09378	0.2533	1660	0.9991
sigma_s_9	0.39170	0.04331	0.31970	0.4893	1832	0.9992
sigma_s_10	0.05930	0.02529	0.03086	0.1258	1536	1.0010
sigma_s_11	0.74980	0.14390	0.53390	1.0830	1670	1.0000
sigma_s_12	0.67300	0.16150	0.44330	1.0220	1626	0.9992
sigma_h_1	0.27860	0.08626	0.16670	0.4797	1667	1.0030
sigma_h_2	0.38630	0.05780	0.29060	0.5138	1688	1.0010
sigma_h_3	0.34160	0.06645	0.23950	0.4949	1775	1.0000
sigma_h_4	0.41350	0.10410	0.26770	0.6684	1653	1.0030
sigma_h_5	0.34810	0.07188	0.24150	0.5215	1675	1.0050
sigma_h_6	0.53970	0.10640	0.37430	0.7816	1694	1.0000
sigma_h_7	0.60680	0.06568	0.49550	0.7437	1728	1.0020
sigma_h_8	0.07534	0.03458	0.03561	0.1616	1460	1.0060
sigma_h_9	0.48020	0.05574	0.38780	0.6140	1722	1.0020

Table 16: (continued)

Parameter	mean	sd	q2.5	q97.5	N_eff	rhat
sigma_h_10	0.63860	0.15130	0.41770	1.0090	1579	0.9997
sigma_h_11	0.51880	0.12210	0.34370	0.8095	1635	1.0010
sigma_h_12	0.33720	0.09447	0.20800	0.5562	1585	1.0000
sigma_c_1	0.75850	0.11560	0.56890	1.0350	1818	1.0030
sigma_c_2	0.51440	0.08485	0.37750	0.7106	1601	0.9999
sigma_c_3	0.46820	0.09498	0.32330	0.6991	1717	0.9990
sigma_c_4	0.53990	0.14470	0.34750	0.8770	1625	1.0020
sigma_c_5	0.80880	0.22570	0.51750	1.3730	1541	1.0010
sigma_c_6	0.53470	0.12420	0.34630	0.8264	1504	1.0020
sigma_c_7	0.54180	0.07147	0.42220	0.6995	1824	1.0010
sigma_c_8	0.45620	0.05773	0.35660	0.5843	1682	1.0000
sigma_c_9	0.55050	0.07468	0.42690	0.7151	1328	1.0060
sigma_c_10	0.76430	0.18330	0.48970	1.1870	1539	1.0030
sigma_c_11	0.51890	0.14430	0.32190	0.8584	1691	1.0010
sigma_c_12	0.44720	0.17250	0.24310	0.9123	1534	1.0030
sigma_l_1	1.15600	0.26350	0.78040	1.7640	1629	1.0010
sigma_l_2	1.41700	0.17410	1.12500	1.8120	1661	1.0020
sigma_l_3	1.48200	0.47540	0.85960	2.7150	1602	1.0010
sigma_l_4	2.81900	1.54700	1.06200	7.2900	1535	0.9996
sigma_l_5	1.44000	0.18910	1.13400	1.8680	1776	1.0020
sigma_l_6	3.04600	1.63600	1.20800	7.7660	1764	1.0020
sigma_l_7	1.18500	0.60260	0.52930	2.7920	1591	0.9998
sigma_l_8	1.66500	0.52390	0.97970	2.9060	1671	0.9997
sigma_l_9	1.94300	0.86160	0.91890	4.2790	1583	1.0050
sigma_l_10	1.07400	0.42720	0.55490	2.0890	1519	1.0030
sigma_l_11	2.21700	2.05900	0.09644	7.9520	1337	1.0040
sigma_l_12	1.17600	0.75710	0.43260	3.2730	1655	0.9995
tau_b	1.00900	0.08510	0.84600	1.1830	1668	1.0010